

$$\equiv x_i + 20 + \left(5\left(\frac{1^2}{x}\right) + 0.00\right) \frac{1.2}{h \bar{c}_j}$$

$$f(x + (5x - x)(0) + 10) = x_i$$

$$2\left(\frac{v}{x}\right)^2 + (x) + \left(\frac{2x}{x}\right) = I_{12}((x)) = \frac{2^2}{h} = 5 \frac{2}{x}$$

$$\frac{x(1-x)^{\frac{2}{3}}}{h x^3}$$

$$f(\bar{x}_2)$$

$$x_k + x\left(\frac{1}{h} + x + \left(\frac{kx}{m^2}\right) \mp 1\right)$$

$$= \frac{x}{x_j} + 0 - (x^2 + 0) = \frac{x}{l_x a m^{x_j}}$$

$$f + (0 + 5) + x_i = (1 = 3)$$

Engineering Mathematics III: Second Order Homogenous ODEs

Week 3 Lecture 3

Housekeeping

1 Pre-class recordings

Available on Canvas - access both sections

2 Assignment 1

Quizzes 1 to 6

Additional component

3 Canvas updates

New material released Fridays

4 Tutorials



Outline – Section 1

Second Order Homogeneous ODEs

Definition and characteristics

Superposition Principle

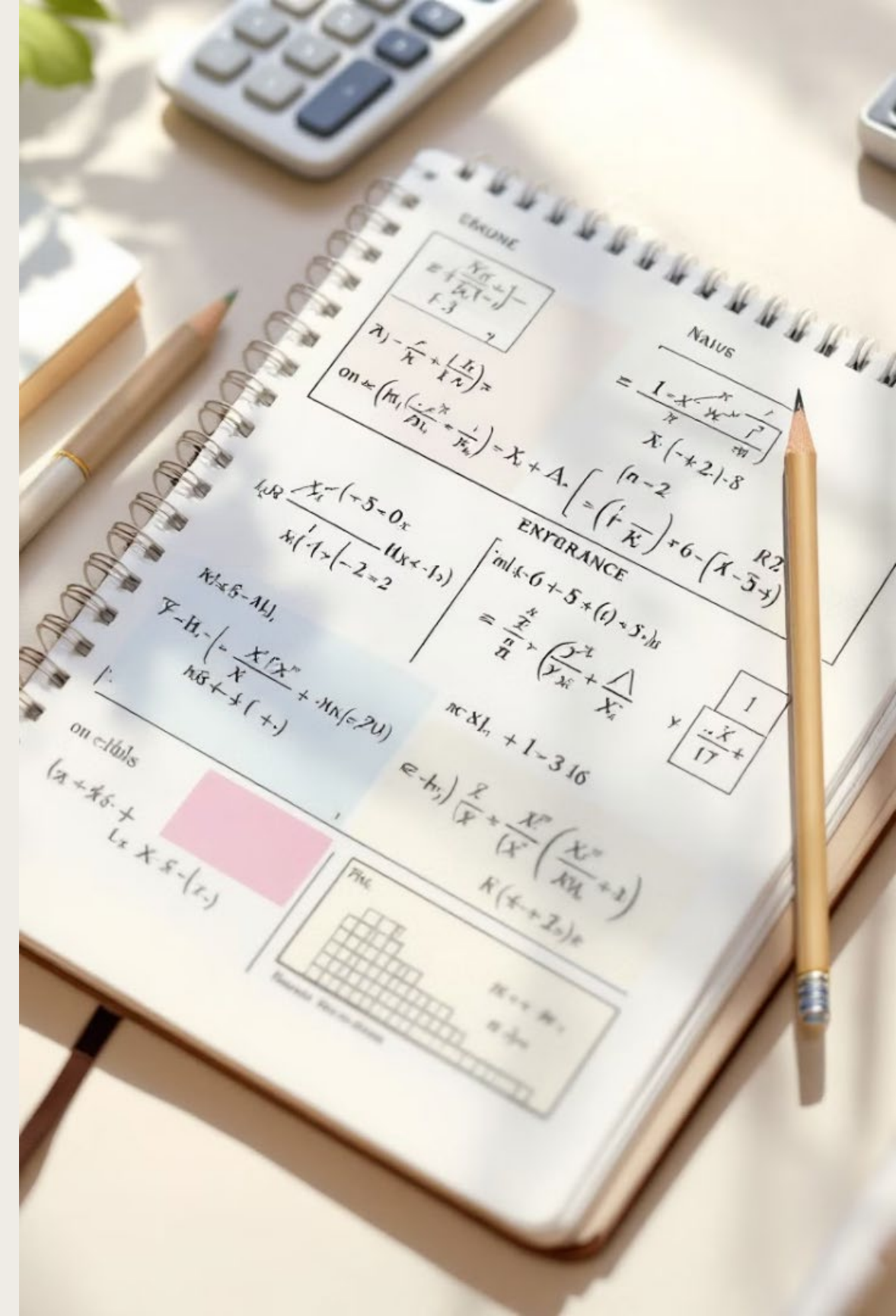
Linear combinations and applications

General Solutions

Form and components

Initial Value Problems

Applying boundary conditions



What are Second Order linear ODEs?

1

Definition

Highest derivative is second derivative

2

Linear ODE

$$y'' + p(x)y' + q(x)y = r(x)$$

3

Homogeneous

When $r(x) = 0$

$$y'' + p(x)y' + q(x)y = 0$$

Physical Applications

Wave mechanics, electrical circuits, and structural analysis in engineering systems.

Homogeneous	Nonhomogeneous

Linear	Nonlinear

Homogeneous

$$y'' + p(x)y' + q(x)y = 0$$

$$y'' + 7y' + 10y = 0$$

$$xy'' + y' + xy = 0$$

Nonhomogeneous

$$y'' + p(x)y' + q(x)y = r(x)$$

$$y'' + 25y' = \cos(x)$$

Linear

Nonlinear

The Superposition Principle

Key Concept

For homogeneous linear ODEs, linear combinations of solutions are also solutions.

ie the sum of two solutions is also a solution.

A scalar multiple is also a solution.

Example

If $y=\sin(t)$ and $y=\cos(t)$ are solutions to $y''+y=0$,

then $y = a \cdot \sin(t) + b \cdot \cos(t)$ is also a solution.

Limitation

Only works for homogeneous ODEs

$$y'' + p(x)y' + q(x)y = 0$$

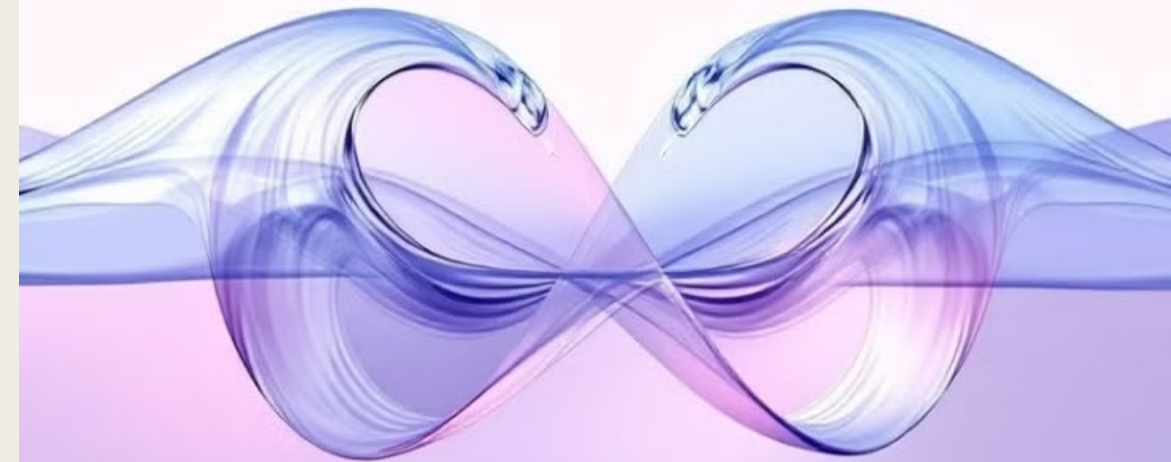


The Superposition Principle

Example

$y=\sin(t)$ and $y=\cos(t)$ are solutions to $y''+y=0$

Check $y=a\cdot\sin(t)+b\cdot\cos(t)$ is also a solution to $y''+y=0$



Non-homogeneous Linear ODE

Example Equation

$$y'' - y = x$$

Consider two particular solutions:

$$y_1 = x \text{ and } y_2 = -1$$

Verification Steps

For $y_1 = x$:

$$y_1'' = 0, \text{ so } 0 - x = x$$

For $y_2 = -1$:

$$y_2'' = 0, \text{ so } 0 - (-1) = 1$$

Both satisfy our equation!

Challenge Question

Is $y_3 = y_1 + y_2 = x - 1$ a solution?

If we check: $y_3'' - y_3 = 0 - (x - 1) = -(x - 1)$

This doesn't equal x , so it's not a solution!

General Solutions for 2nd order homogeneous linear ODEs

General solution form

$$y = c_1 y_1 + c_2 y_2$$

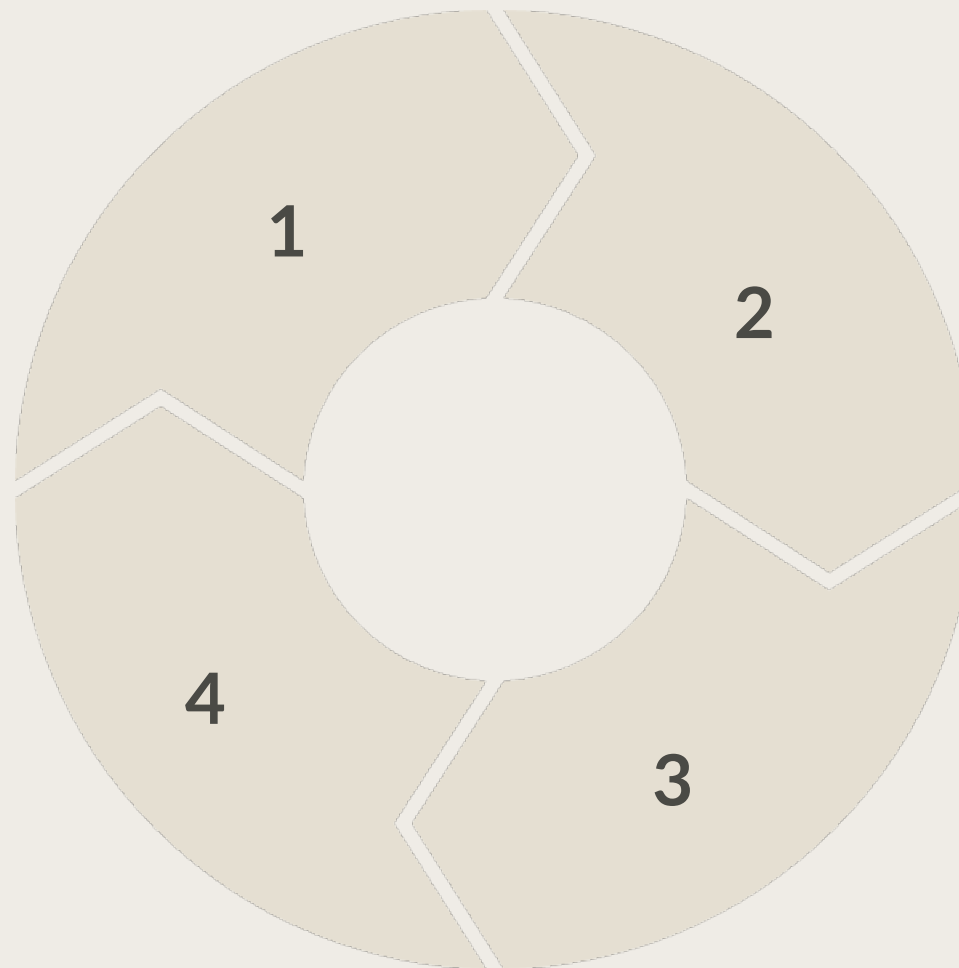
As superposition principle applies

Linearly Independent?

Can be complicated.

For a 2nd order ODE:

Solutions aren't scalar multiples
of each other.



Components

y_1, y_2 are linearly independent
solutions of the ODE.

c_1, c_2 are arbitrary constants.

Constants

c_1, c_2 determined by initial conditions.

2nd Order Linear ODE

Initial Value Problems



Two initial conditions

$y(t_0)$ and $y'(t_0)$

2

Find Solutions

Find two linearly independent solutions y_1 and y_2
- *can be tricky. Can be given.*

3

Find the General Solution

$y = c_1 y_1 + c_2 y_2$ - easy once you have found y_1 and y_2

4

Apply Conditions

Use $y(t_0)$ and $y'(t_0)$ to find c_1 and c_2



Example: Solving $y'' + y = 0$

Given

Initial conditions: $y(0) = 1$, $y'(0) = 0$

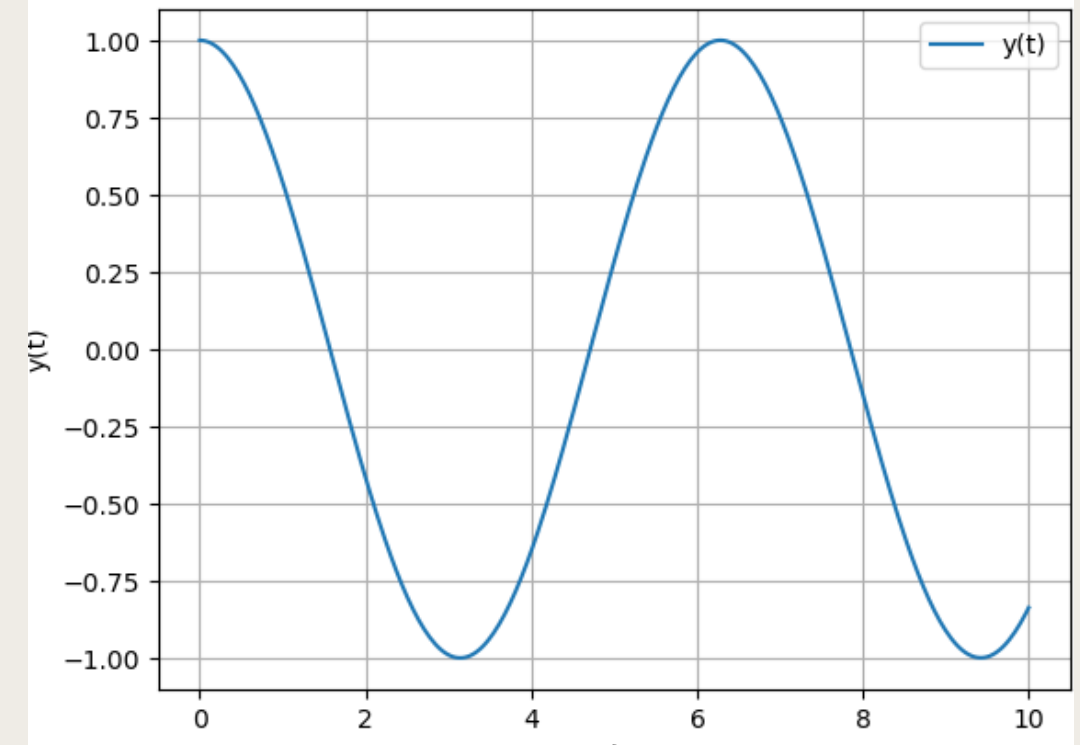
$y_1 = \sin(t)$ and $y_2 = \cos(t)$

General Form

$$y = c_1 \sin(t) + c_2 \cos(t)$$

Final Solution

$$y = \cos(t)$$



Example: Solving $y'' + y = 0$

Given

Initial conditions: $y(0) = 1$, $y'(0) = 0$

$y_1 = \sin(t)$ and $y_2 = \cos(t)$

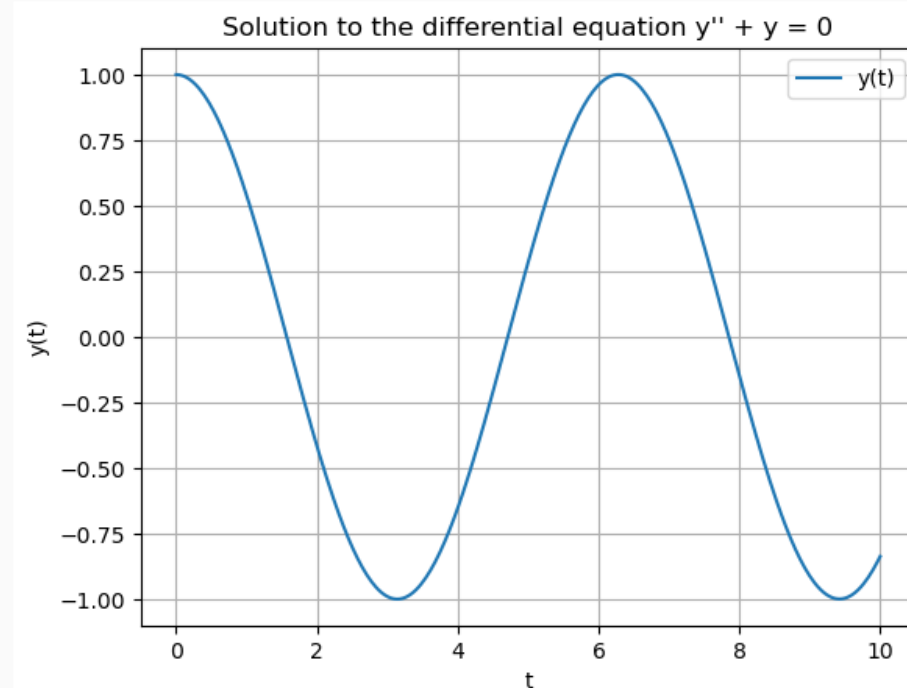
Step-by-Step Solution:

1. **Rewrite the differential equation:** The given differential equation is $y'' + y = 0$.
2. **Solve the characteristic equation:** The characteristic equation for this differential equation is: $r^2 + 1 = 0$
Solving for r , we get: $r = \pm i$ So, the general solution to the differential equation is: $y(t) = C_1 \cos(t) + C_2 \sin(t)$
3. **Apply the initial conditions:**
 - For $y(0) = 1$: $y(0) = C_1 \cos(0) + C_2 \sin(0)$ $1 = C_1 \cdot 1 + C_2 \cdot 0$ $C_1 = 1$
 - For $y'(0) = 0$: First, find $y'(t)$: $y'(t) = -C_1 \sin(t) + C_2 \cos(t)$ Apply the initial condition: $y'(0) = -C_1 \sin(0) + C_2 \cos(0)$ $0 = -1 \cdot 0 + C_2 \cdot 1$ $C_2 = 0$
4. **Write the particular solution:** Substituting C_1 and C_2 into the general solution, we get: $y(t) = \cos(t)$

Final Solution

$$y = \cos(t)$$

Here is the graph of the solution $y(t) = \cos(t)$:



Practice Exercise

1

Problem

$$y'' - 3y' + 2y = 0, \text{ with } y(0) = 0, y'(0) = 1$$

2

Given Solutions

$$y_1 = e^x \text{ and } y_2 = e^{2x}$$

3

General Solution

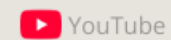
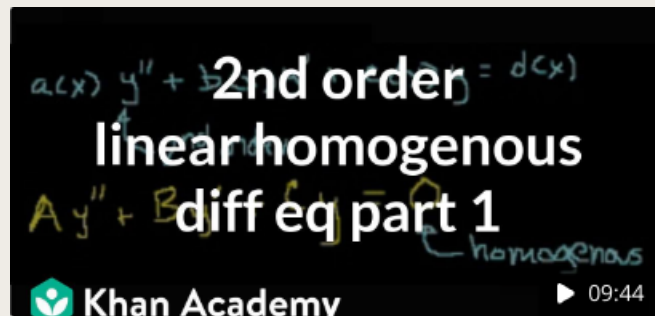
$$y = c_1 e^x + c_2 e^{2x}$$

IVP Solution

Work through to find c_1 and c_2 using the initial conditions.

$$y(x) = (e^x - 1)e^x$$

Textbook: Chapter 2.1

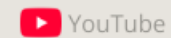
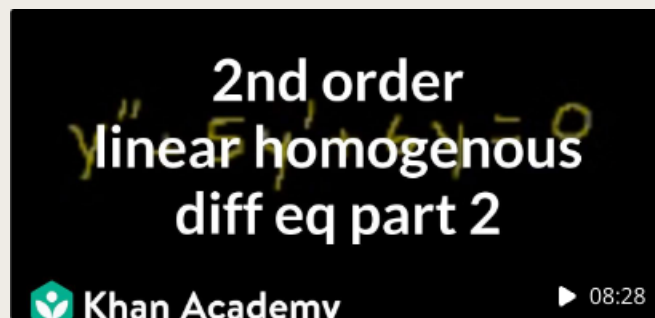


YouTube



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2nd order linear homogeneous differential equations 2 | Khan Acade...

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5

Solve the Characteristic Equation

$$\lambda^2 + a\lambda + b = 0.$$

Using the quadratic formula we get the following solution:

$$\lambda_1 = \frac{1}{2}(-a + \sqrt{a^2 - 4b}) \text{ and } \lambda_2 = \frac{1}{2}(-a - \sqrt{a^2 - 4b})$$

Giving solutions

$$y_1 = e^{\lambda_1 x} \text{ and } y_2 = e^{\lambda_2 x}$$

6

The General Solution

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

A version of the Quadratic Formula

$$\text{If } \lambda^2 + a\lambda + b = 0$$

$$\text{then } \lambda = \frac{-a \pm \sqrt{a^2 - 4b}}{2}$$

Two real solutions

Condition

When $a^2 - 4b > 0$, the characteristic equation has two distinct real roots λ_1 and λ_2 .

Solutions

The two linearly independent solutions are:

$$y_1 = e^{\lambda_1 x} \text{ and } y_2 = e^{\lambda_2 x}$$

General Solution

The general solution is $y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$, where c_1 and c_2 are arbitrary constants.

This standard form has a coefficient of 1 in front of y'' .

Outline

1 The Solution Method

Using the characteristic equation approach to find solutions to 2nd order Homogeneous ODEs

2 Case 1: Two Real Solutions

When the characteristic equation has distinct real roots

3 Case 2: One Real Root

When the characteristic equation has a repeated real root

4 Case 3: Complex Conjugate Roots

When the characteristic equation has complex roots

5 Summary and Applications

Reviewing all solution methods and their applications

The Solution Method

1

Start with an Assumption

In our first lecture we showed that

$y = e^{-kx}$ is the **solution to $y' + ky = 0$** ,
for some constant k .

2

Find the Derivatives

It could be reasonable to 'assume' $y = e^{\lambda x}$ is a **solution to $y'' + ay' + by = 0$** for some constant λ .

Then we can calculate

$y' = \lambda e^{\lambda x}$ and $y'' = \lambda^2 e^{\lambda x}$.

3

Substitute into Original Equation

Substituting these values into our original equation $y'' + ay' + by = 0$,

we get $\lambda^2 e^{\lambda x} + a\lambda e^{\lambda x} + be^{\lambda x} = 0$

$$e^{\lambda x}(\lambda^2 + a\lambda + b) = 0.$$

4

The Characteristic Equation

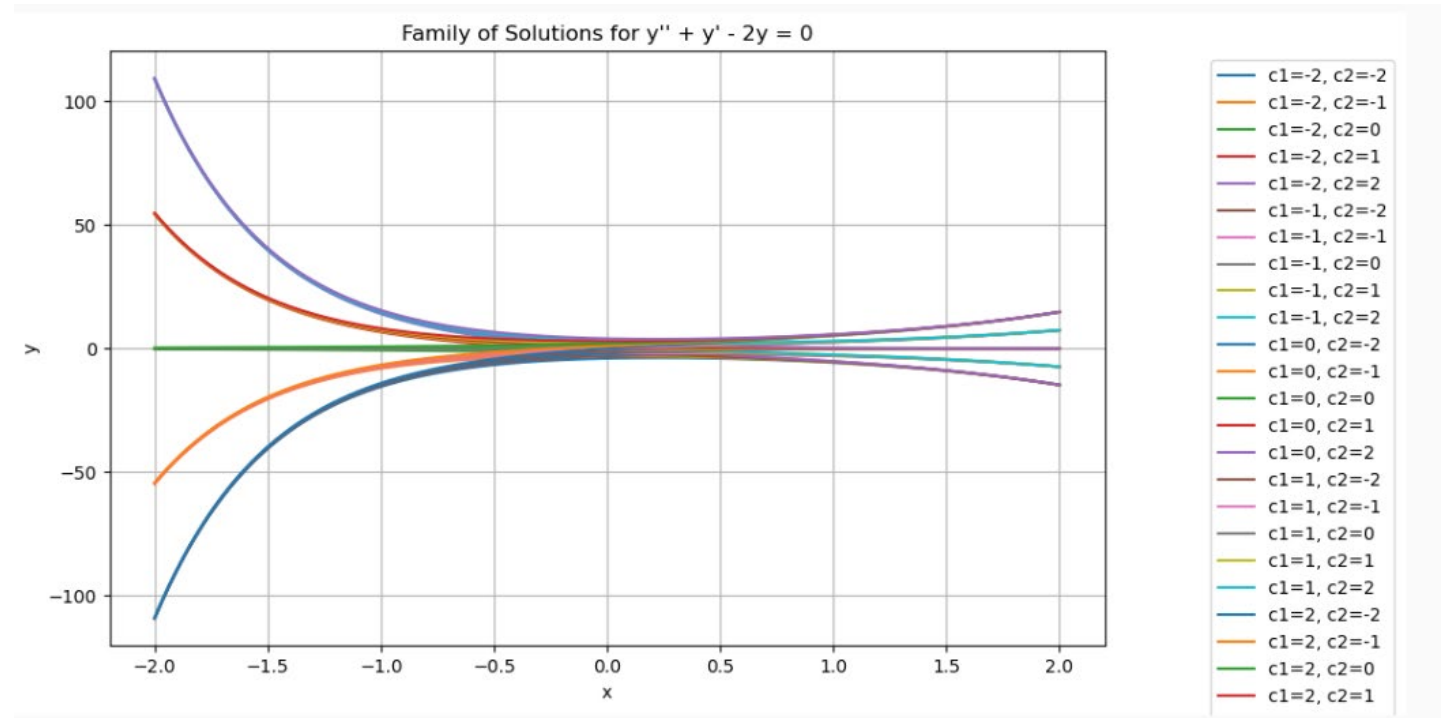
Since $e^{\lambda x}$ is never 0, we must have

$$\lambda^2 + a\lambda + b = 0.$$

This is called the **characteristic equation** of its corresponding differential equation, which we solve using the quadratic formula.

Example: For the equation $y'' + y' - 2y = 0$, the characteristic equation is $\lambda^2 + \lambda - 2 = 0$, which gives $\lambda_1 = 1$ and $\lambda_2 = -2$.

Therefore, the general solution is $y = c_1 e^x + c_2 e^{-2x}$



Case 2: One Real Root

1

Condition

When $a^2 - 4b = 0$, the characteristic equation has one repeated real root λ .

2

First Solution

We have one solution $y_1 = e^{\lambda_1 x}$.

However, we need another.

3

Second Solution

In this case it is also true that $y_2 = xe^{\lambda_2 x}$ is a solution. **Check!**

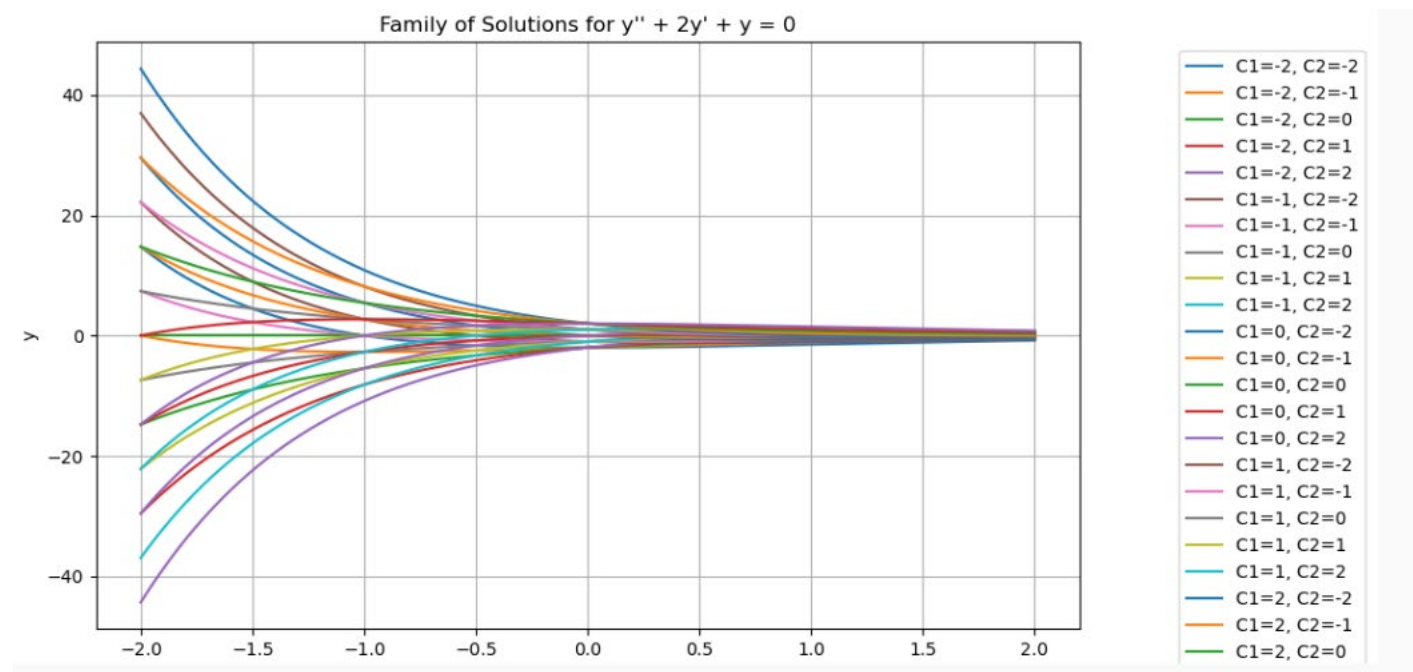
The 1st and 2nd solution are linearly independent (ie they are not proportional).

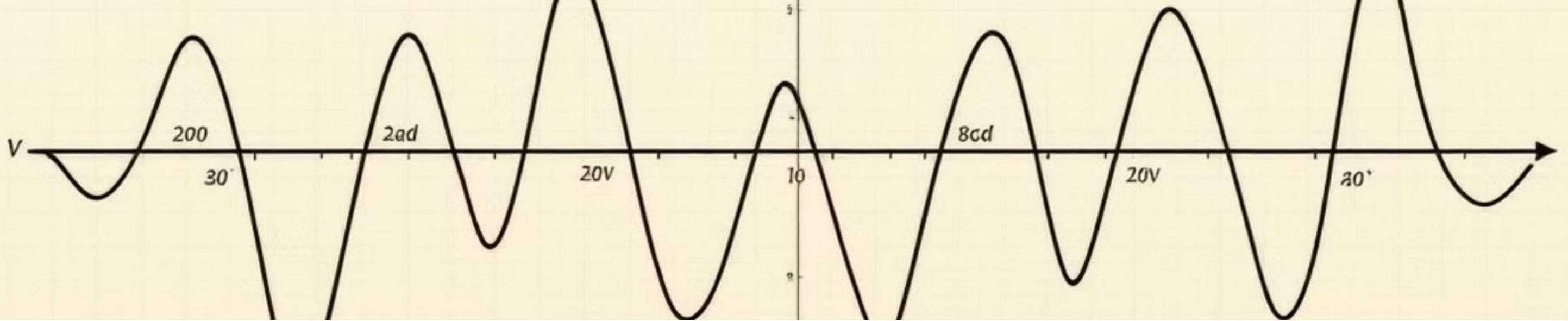
4

General Solution

The general solution is $y = c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_2 x}$,
where c_1 and c_2 are arbitrary constants.

Exercise: For the equation $y'' + 2y' + y = 0$, find the characteristic equation and determine the general solution.





Case 3: Complex Conjugate Roots

1 Condition

When $a^2 - 4b < 0$, the characteristic equation has two complex conjugate roots $\lambda = -a/2 \pm i\omega$, where $\omega^2 = b - (a^2/4)$.

2 Solutions

The linearly independent solutions are

$$y_1 = e^{\frac{-ax}{2}} \cos(\omega x)$$

and

$$y_2 = e^{\frac{-ax}{2}} \sin(\omega x).$$

3 General Solution

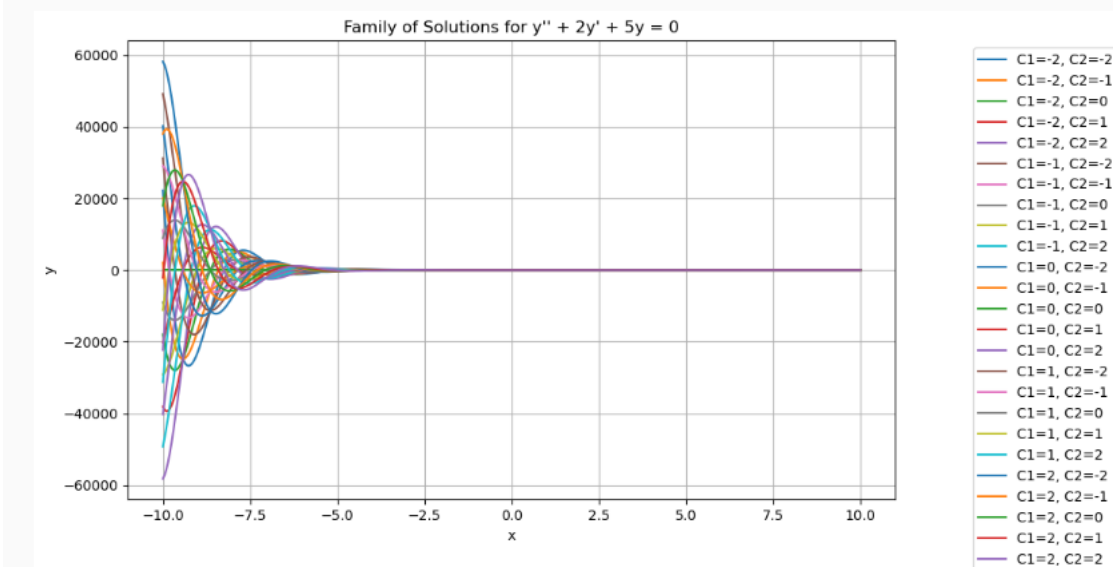
The general solution is

$$y = e^{\frac{-ax}{2}} (A \cos(\omega x) + B \sin(\omega x)),$$

where A and B are arbitrary constants.

Example: For the equation $y'' + 2y' + 5y = 0$, the characteristic equation has roots $-1 \pm 2i$.

So $\omega = 2$, and the solution is $y = e^{-x}(\text{Acos}(2x) + \text{Bsin}(2x))$.



To solve the differential equation $y'' + 2y' + 5y = 0$, we first find the characteristic equation and then determine the general solution.

1. **Characteristic Equation:** The given differential equation is: $y'' + 2y' + 5y = 0$

To find the characteristic equation, we assume a solution of the form $y = e^{rx}$. Substituting this into the differential equation, we get: $r^2 e^{rx} + 2r e^{rx} + 5e^{rx} = 0$ Dividing through by e^{rx} (which is never zero), we obtain the characteristic equation: $r^2 + 2r + 5 = 0$

2. **Solving the Characteristic Equation:** The characteristic equation is a quadratic equation: $r^2 + 2r + 5 = 0$

Solving for r using the quadratic formula $r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, we get: $r = \frac{-2 \pm \sqrt{4 - 20}}{2} = \frac{-2 \pm \sqrt{-16}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i$

3. **General Solution:** Since we have complex roots $r = -1 \pm 2i$, the general solution to the differential equation is: $y(x) = e^{-x}(C_1 \cos(2x) + C_2 \sin(2x))$

So, the general solution to the differential equation $y'' + 2y' + 5y = 0$ is: $y(x) = e^{-x}(C_1 \cos(2x) + C_2 \sin(2x))$

Example: Second-Order Homogeneous ODE

Find the general solution for:

$$y'' - 4y' + 4y = 0$$

Find Characteristic Equation

$$r^2 - 4r + 4 = 0$$

Solve for Roots

$$(r - 2)^2 = 0, \text{ so } r = 2 \text{ (repeated root)}$$

Write Solutions

$$y_1 = e^{2x} \text{ and } y_2 = xe^{2x}$$

Express General Solution

$$y = c_1e^{2x} + c_2xe^{2x}$$

Example 3: $y'' + 4y' + 5y = 0$

For this equation, the characteristic equation is $\lambda^2 + 4\lambda + 5 = 0$. Find the roots and determine which case applies, then write the general solution.



Summary of Solution Methods

Case	Roots	General Solution
1	Distinct real roots λ_1, λ_2	$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$
2	One real root λ	$y = (c_1 + c_2 x) e^{\lambda x}$
3	Complex conjugate roots $-a/2 \pm i\omega$ where $\omega^2 = b - (a^2/4)$.	$y = e^{(-ax/2)} (A \cos(\omega x) + B \sin(\omega x))$

The solution method for second order homogeneous ODEs with constant coefficients can be summarized in these three cases.

The type of roots in the characteristic equation determines which formula to use for the general solution.

Key Takeaways

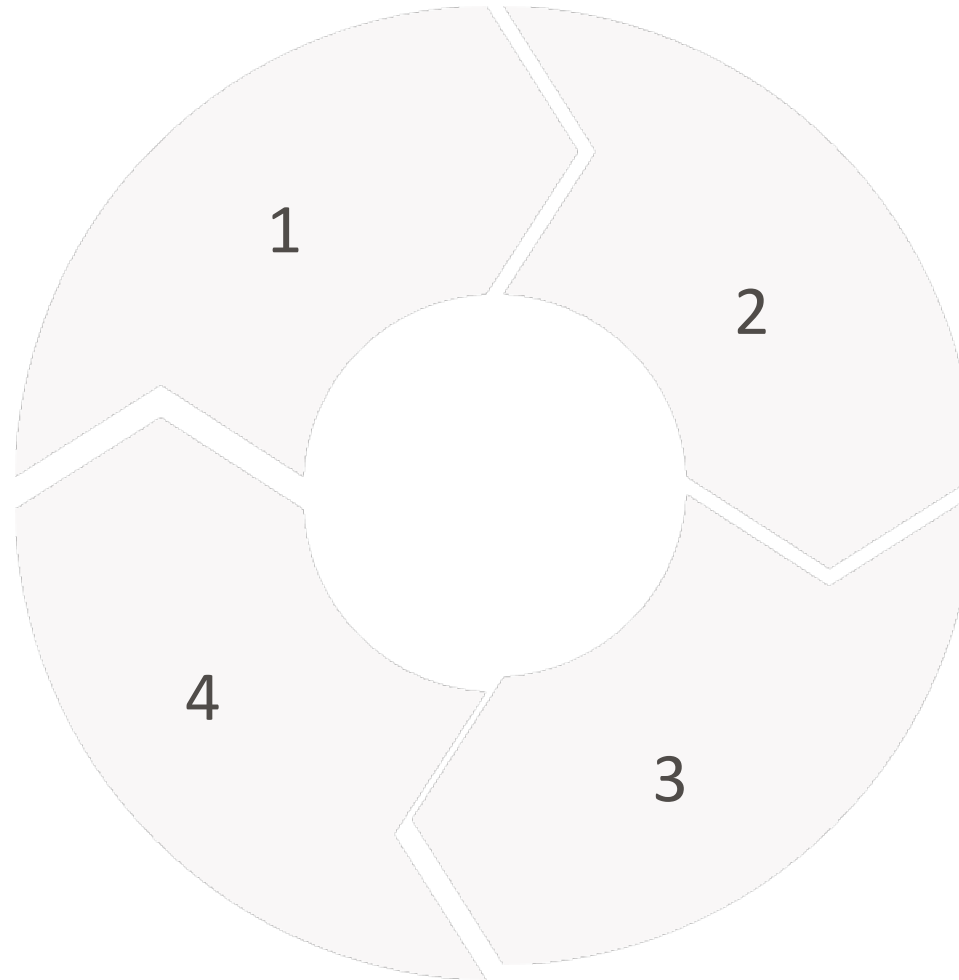
Identify the Equation Form

Ensure the 2nd order homogeneous linear ODE is in standard form $y'' + ay' + by = 0$ with constant coefficients.

Apply the Appropriate Formula

Formula

Base on the type of roots found, use the corresponding solution formula to find the general solution..



Find the Characteristic Equation Equation

Assume $y = e^{\lambda x}$ is a solution.
Write and solve $\lambda^2 + a\lambda + b = 0$ to determine the roots.

Determine the Case

Check if $a^2 - 4b$ is positive, zero, or negative to identify which case applies.

Remember to refer to **Chapter 2.2 in your textbook** for additional details and examples.

Example 2: $y'' + 4y' + 4y = 0$

For this equation, the characteristic equation is $\lambda^2 + 4\lambda + 4 = 0$. Find the roots and determine which case applies, then write the general solution.



Practice Examples

Example 1: $y'' + 4y' + 3y = 0$

For this equation, the characteristic equation is $\lambda^2 + 4\lambda + 3 = 0$.

Find the roots and determine which case applies (two real roots, one repeated root or complex roots), then write the general solution.

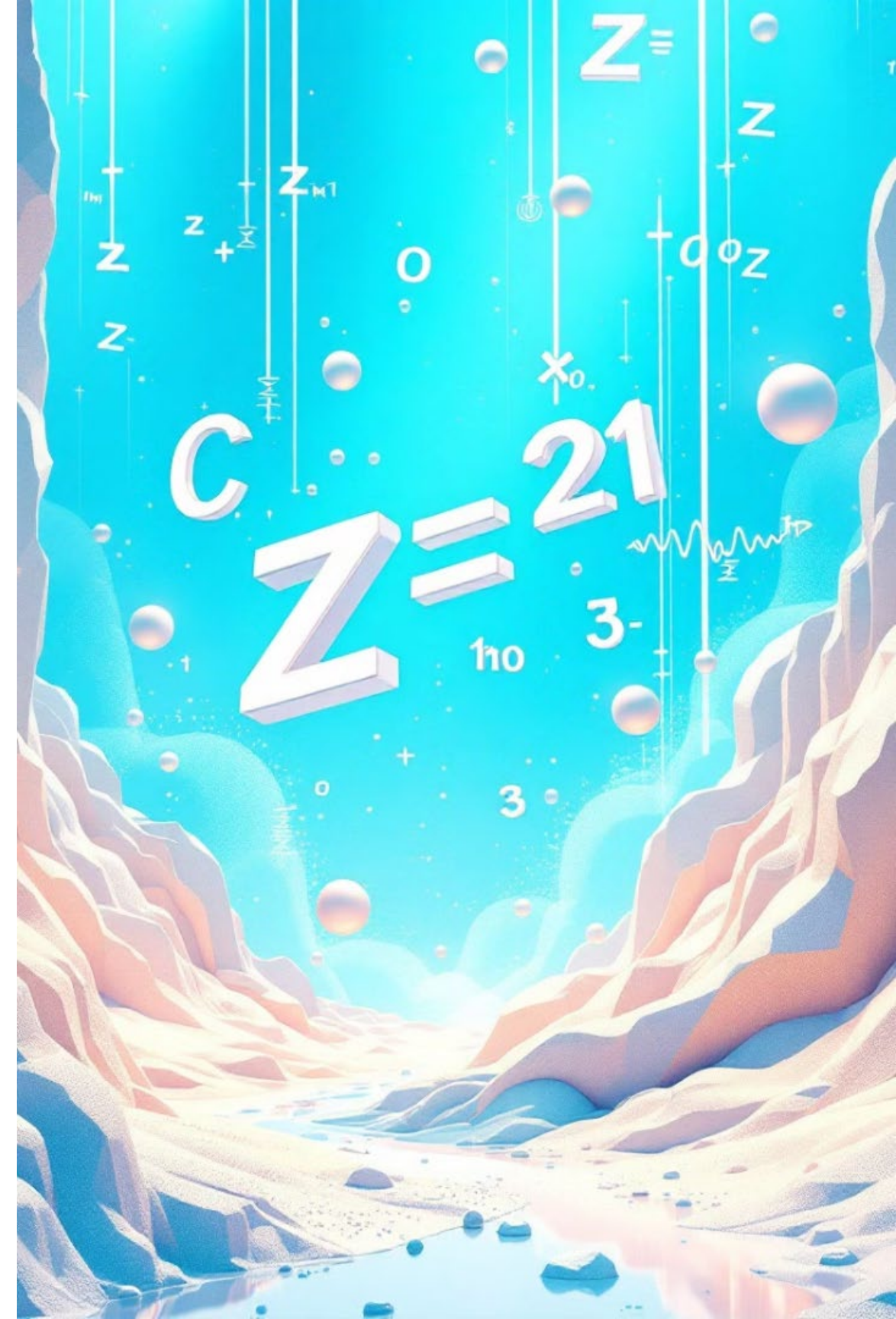
Application of Second-Order Understanding Vibration

Second order linear homogeneous ODEs with constant coefficients are part of a number of fundamental physical models.

In particular 2nd order ODEs govern vibration and many other periodic phenomena.

The mathematics involved is essential to modelling and understanding many mechanical, electrical, and other types of physical phenomena.

Vibration and the Harmonic Oscillator



The 2010 Chilean Earthquake: A Case Study in Structural Vibration

On February 27, 2010 at 3:34 a.m., an a.m., an earthquake of magnitude magnitude 8.8 struck Chile, one of the strongest quakes ever recorded. The shaking lasted for more than two minutes and when it when it was over at least 500 people people had been killed. Damage was was estimated to be in excess of \$30,000,000,000.

What makes this event particularly particularly noteworthy is the relatively low death toll given the the earthquake's magnitude. By comparison, a 7.0 earthquake in Haiti just one month earlier resulted resulted in an estimated 100,000-100,000-250,000 fatalities

One of the primary factors for the the lower death toll in the Chilean Chilean earthquake are the strict building codes that exist in that country and their stringent enforcement. These building codes codes were adopted after a massive massive quake of magnitude 9.5 there in 1960, the strongest earthquake ever recorded.

1960

1

Chile experiences 9.5 magnitude earthquake (strongest ever recorded) leading to adoption of stringent building codes

2

January 10, 2010

Haiti struck by 7.0 magnitude earthquake resulting in 100,000-250,000 deaths due largely to poor construction standards

3

February 27, 2010

8.8 magnitude earthquake hits Chile with relatively low death toll (500) thanks to earthquake-resistant building design

The difference?

Chile's strict building codes implemented after their devastating 1960 earthquake. These regulations, enforced by a law holding builders liable for damages for ten years post-construction, mandated earthquake-resistant designs.

Understanding Vibration in Engineering

Building Codes

For ten years after a building's construction, builders in Chile are held liable for any damages that result from not adhering to these codes. The high death toll in Haiti has been attributed in large part to the poor construction of many buildings and lack of an enforced code that mandates structures be resistant to earthquakes.

Mathematical Foundation

Much of the relevant mathematics falls into the realm of realm of second-order differential equations. This mathematics governs not only mechanical vibration, but vibration, but many electromagnetic and other physical physical phenomena.

1

2

3

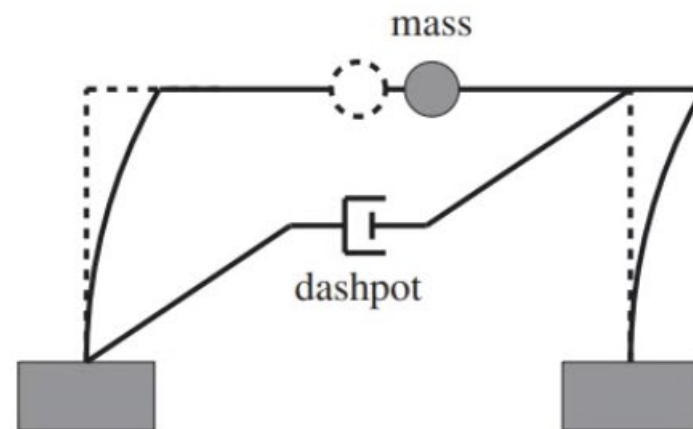
Engineering Importance

A detailed understanding of how mechanical objects vibrate is at the heart of engineering structures that can endure this kind of abuse. Such an understanding allows engineers to design buildings that can respond to earthquakes without collapsing or endangering occupants.

The Harmonic Oscillator: Simplified 2-D model of a building

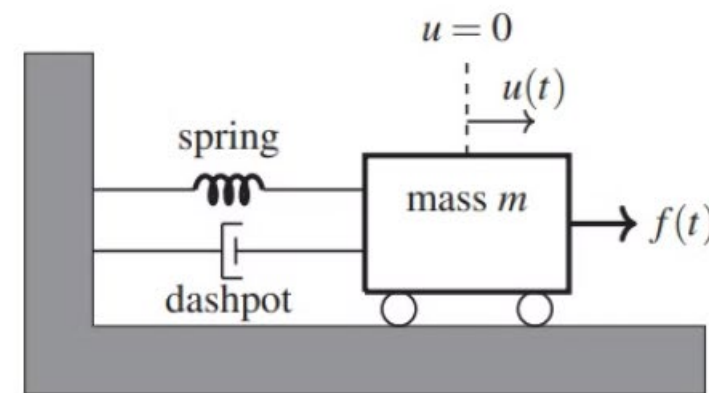
Building Model

Consider a highly simplified two-dimensional dimensional model of a building. It consists of a single story, with the roof depicted as a point mass m supported by by vertical walls. When the mass is displaced displaced from its equilibrium position, the the walls exert a force that opposes this displacement, and the magnitude of the force the force is proportional to the magnitude of magnitude of the displacement.



Abstraction

A simplified abstraction of the situation is shown in which the roof mass is the cart on wheels and the walls' restoring force is embodied by the spring. Motion of the roof is also opposed by frictional forces with magnitude in proportion to the speed of the mass; this is embodied by the dashpot (or damper).



Left panel: a simplified one-story building. Right panel: a mass-spring-damper abstraction of such a building.

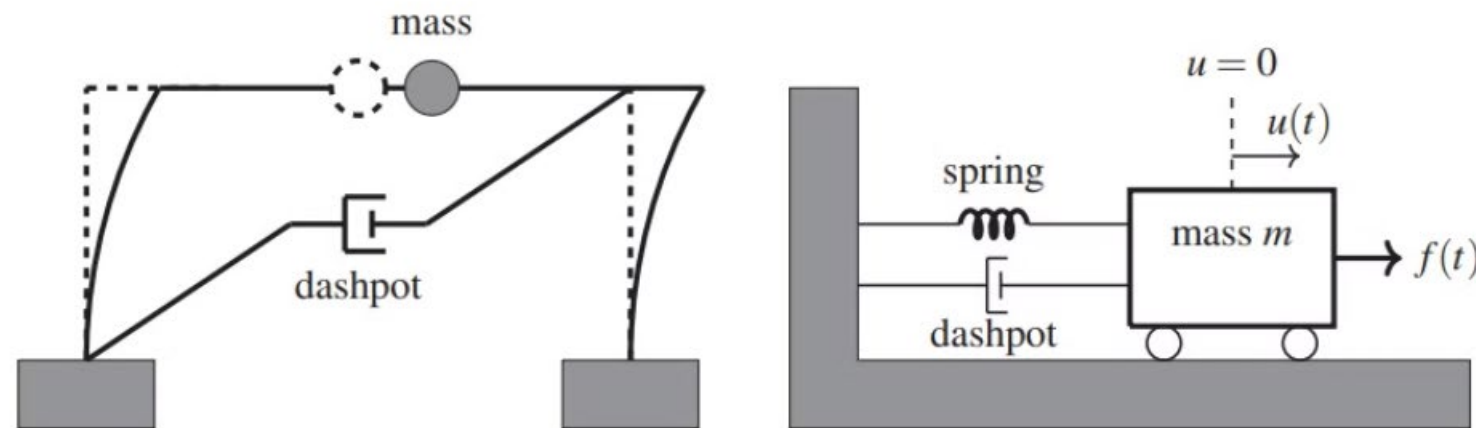
The Harmonic Oscillator

Earthquake Modeling

An earthquake might be modeled as an additional externally applied force $\mathbf{f(t)}$ on the mass. Our interest here is in the lateral back and forth motion of the mass m as a function of time.

Let $\mathbf{u(t)}$ denote the horizontal displacement at time t of the mass in the right panel from its **equilibrium position**.

Take $u > 0$ to indicate displacement to the right. We will assume the spring and dashpot (damper) have negligible mass.



Left panel: a simplified one-story building. Right panel: a mass-spring-damper abstraction of such a building.

Hooke's Law and Damping Forces

Hooke's Law

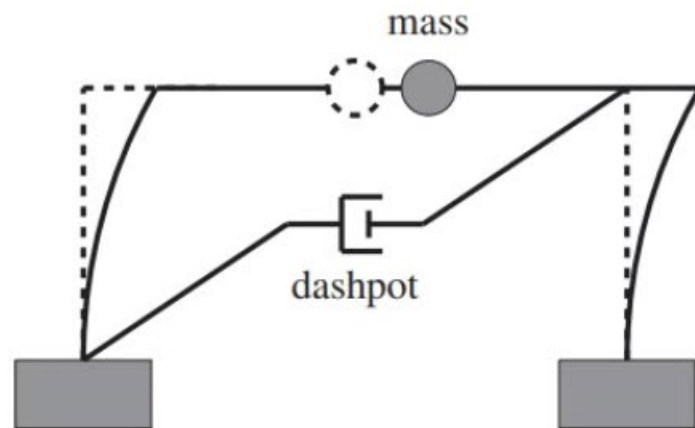
The assumption that the spring exerts a force in proportion to and opposed to the displacement u is known as Hooke's law.

Hooke's law is known as

$$F_{\text{spring}} = -ku \quad (4.1)$$

where k is the nonnegative spring constant and u is displacement.

The negative sign indicates the force opposes displacement.



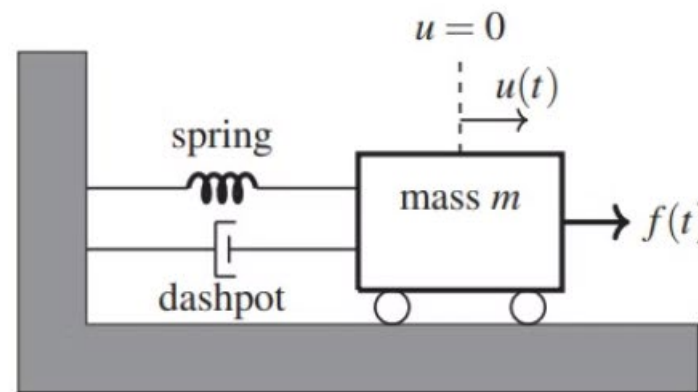
Damping Force

The dashpot acts as a frictional or **damping** force of the form

$$F_{\text{damping}} = -c u' \quad (4.2)$$

where c is the nonnegative damping constant and u' is velocity. This viscous damping is always opposed to the direction of motion.

The linear relation between force and velocity in (4.2) is known as viscous damping.



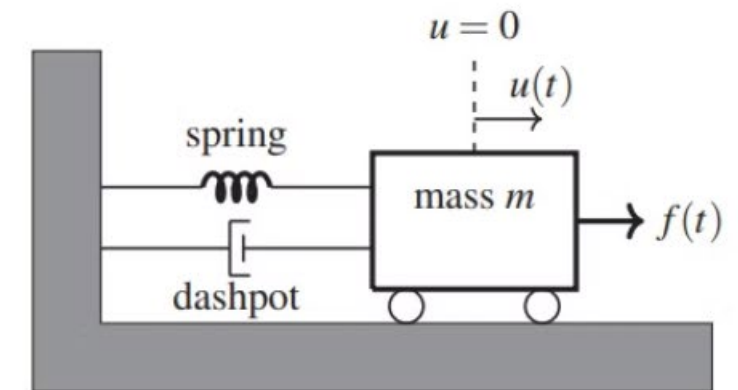
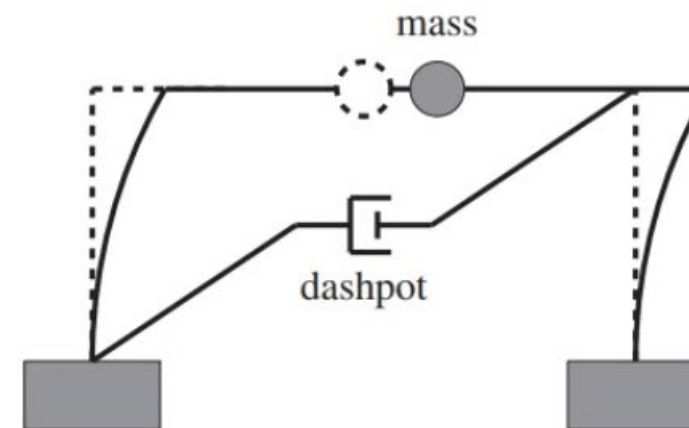
Reading Exercise 4.1.1 Use Newton's second law of motion, $F = ma$, along with (4.1) and (4.2), to write down a relation between m , c , k , u , u' , u'' , and $f(t)$. Here a is the horizontal acceleration of the mass m and F denotes the net horizontal force on m , the sum of the spring and dashpot forces and the external force $f(t)$.

$$F_{\text{spring}} = -ku \quad (4.1)$$

$$F_{\text{damping}} = -c u' \quad (4.2)$$

$$F = ma$$

$$mu''(t) + cu'(t) + ku(t) = f(t),$$



We have $F_{\text{spring}} = -ku$ and $F_{\text{damping}} = -cu'$, along with external force $f(t)$ on the mass. The net force is thus $F = -ku - cu' + f(t)$, so from $F = ma$ and $a = u''$ this yields $mu'' = -ku - cu' + f(t)$ or $mu'' + cu' + ku = f(t)$.

When combined with Newton's second law ($F = ma$), these principles yield the **second-order differential equation that governs vibration: $mu''(t) + cu'(t) + ku(t) = f(t)$** , where $f(t)$ represents any external forces like earthquakes.

Finally, Newton's law says that the sum of the forces acting on the mass is equal to the mass times acceleration, so

$$F = my''$$

Here F is the sum of all forces, so $F = F_s + F_d$, or

$$my'' = -cy' - ky,$$

rearranging we get

$$my'' + cy' + ky = 0.$$

This is a second order linear homogeneous ODE with constant coefficients, and we will look at the solutions for various values of m , c , and k .

Note in our model that $m, k > 0$ and $c \geq 0$.

Harmonic Oscillator Equations

$$mu''(t) + cu'(t) + ku(t) = f(t) \quad (4.3)$$

Forced Harmonic Oscillator

$mu''(t) + cu'(t) + ku(t) = f(t)$ is the equation of the forced harmonic oscillator, or driven harmonic oscillator.

Unforced Harmonic Oscillator

When there is no external force $f(t)$, so only the spring and dashpot (damper) exert force on m , the equation becomes

$$mu''(t) + cu'(t) + ku(t) = 0 \quad (4.4)$$

This ODE is homogeneous, since the right side is zero, while the forced equation is nonhomogeneous if $f(t)$ is not the zero function.

Undamped Harmonic Oscillator

In the case where $c = 0$, so the system has no frictional forces or damping, the only force acting on the mass is the spring, and the equation becomes

$$mu''(t) + ku(t) = 0 \quad (4.5)$$

This is the undamped harmonic oscillator or pure harmonic oscillator. In this case common sense suggests that if set in motion, the mass should oscillate and never stop.

Case 1: No Damping

Example

Suppose we have a mass - spring system with mass of 4 and spring constant of 16. The displacement at $t = 0$ is 1 and the velocity at $t = 0$ is 2. Find a function that describes the position as a function of time.

Case 1: No Damping

Example

Suppose we have a mass - spring system with mass of 4 and of 4 and spring constant of 16. The displacement at $t = 0$ is 1 and the velocity at $t = 0$ is 2. Find a function that describes the position as a function of time.

Solution

First we know that $\omega = \sqrt{k/m} = \sqrt{16/4} = 2$.

So our general solution is $y = A\cos(2t) + B\sin(2t)$.

Differentiating this gives $y' = -2A\sin(2t) + 2B\cos(2t)$.

Our initial conditions give $y(0) = 1$ and $y'(0) = 2$.

So we have $1 = A\cos(2.0) + B\sin(2.0)$.

$$2 = -2A\sin(2.0) + 2B\cos(2.0).$$

Solving this gives $A = 1$ and $B = 1$

so our solution is $y = \cos(2t) + \sin(2t)$.

Case 1: No Damping

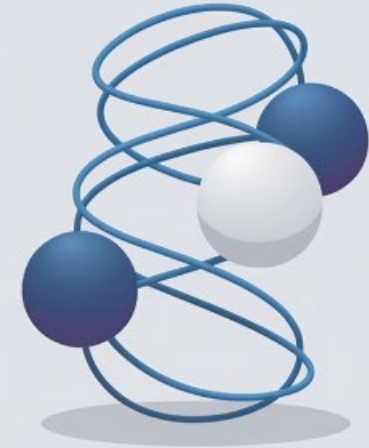
In this case our differential equation is $my'' + ky = 0$

Since $m, k > 0$, we can put this in standard form and solve the characteristic equation $\lambda^2 + k/m = 0$

This has solutions $\lambda = \pm\sqrt{k/m}i$

So the system has angular frequency $\omega = \sqrt{k/m}$

Since $\frac{a}{2}=0$, the general solution will be $y = A\cos(\omega t) + B\sin(\omega t)$



Sure! Let's solve the characteristic equation step by step:

1. **Write down the characteristic equation:** $r^2 + \frac{k}{m} = 0$
2. **Isolate r^2 :** $r^2 = -\frac{k}{m}$
3. **Solve for r** by taking the square root of both sides: $r = \pm\sqrt{-\frac{k}{m}}$
4. **Simplify the expression:** Since $\sqrt{-1} = i$ (where i is the imaginary unit), we get: $r = \pm i\sqrt{\frac{k}{m}}$

So, the roots of the characteristic equation are: $r = i\sqrt{\frac{k}{m}}$ and $r = -i\sqrt{\frac{k}{m}}$

These are purely imaginary roots, which means the general solution to the differential equation will be in terms of sine and cosine functions:

$$y(t) = C_1 \cos\left(\sqrt{\frac{k}{m}}t\right) + C_2 \sin\left(\sqrt{\frac{k}{m}}t\right)$$

Here, C_1 and C_2 are arbitrary constants determined by initial conditions.

Damped Systems

Now lets see what happens when we add a damper to the system. Our differential equation is now:

$$my'' + cy' + ky = 0.$$

As we saw earlier, the solution type will vary depending on the roots of the characteristic equation.

The roots of the characteristic equation come from the quadratic formula, and are given by

$$\lambda_1 = -\alpha + \beta \text{ and } \lambda_2 = -\alpha - \beta$$

$$\text{where } \alpha = c/(2m) \text{ and } \beta = (1/2m)\sqrt{c^2 - 4mk}$$

1 Case 1: Overdamping

In this case $c^2 > 4mk$, and there are two distinct real roots.

2 Case 2: Critical damping

In this case $c^2 = 4mk$, and there is one real root.

3 Case 3: Underdamping

In this case $c^2 < 4mk$, and there are complex conjugate roots.

$$mu''(t) + ku(t) = 0 \quad (4.5)$$

Reading Exercise 4.1.2

- (a) Consider (4.5) when $m = 1$ kg and $k = 1$ newton per meter, so the ODE is $u''(t) + u(t) = 0$. Show that in this case $u(t) = A \cos(t) + B \sin(t)$ provides a solution for any choice of A and B .
- (b) Show that in the general case for (4.5) the function $u(t) = A \cos(\omega t) + B \sin(\omega t)$ with $\omega = \sqrt{k/m}$ provides a solution for any choice of A and B . What is the dimension of ω ?

▼ Answer.

(a) If $u(t) = A \cos(t) + B \sin(t)$ then $u''(t) = -A \cos(t) - B \sin(t)$ and all terms cancel in $u'' + u$, for any choice of A and B , so $u''(t) + u(t) = 0$.

(b) More generally if $u(t) = A \cos(\omega t) + B \sin(\omega t)$ then $u''(t) = -A\omega^2 \cos(\omega t) - B\omega^2 \sin(\omega t)$. In $mu'' + ku = 0$ this becomes

$$A(k - m\omega^2) \cos(\omega t) + B(k - m\omega^2) \sin(\omega t) = 0.$$

If $\omega = \sqrt{k/m}$ the left side above is identically zero and so the equation is satisfied for all t . Given that ωt appears as an argument to sine or cosine, we must have $[\omega] = T^{-1}$. This also follows from $[\omega] = [k]^{1/2}[m]^{-1/2} = T^{-1}$. Thus ω has the dimension of a frequency.

Cases 2 & 3: Critical Damping and Underdamping

Case 2: Critical Damping

Here the general solution is $y = (c_1 + c_2 t)e^{-\alpha t}$.

This is the 'borderline' case between oscillatory and non-oscillatory motion. The solution looks much like the overdamped case graphically. It can be thought of as getting to equilibrium 'as fast as possible'.

Case 3: Underdamping

This is the most complicated case. Here our solutions are

$$\lambda_1 = -\alpha + i\omega \text{ and } \lambda_2 = -\alpha - i\omega,$$

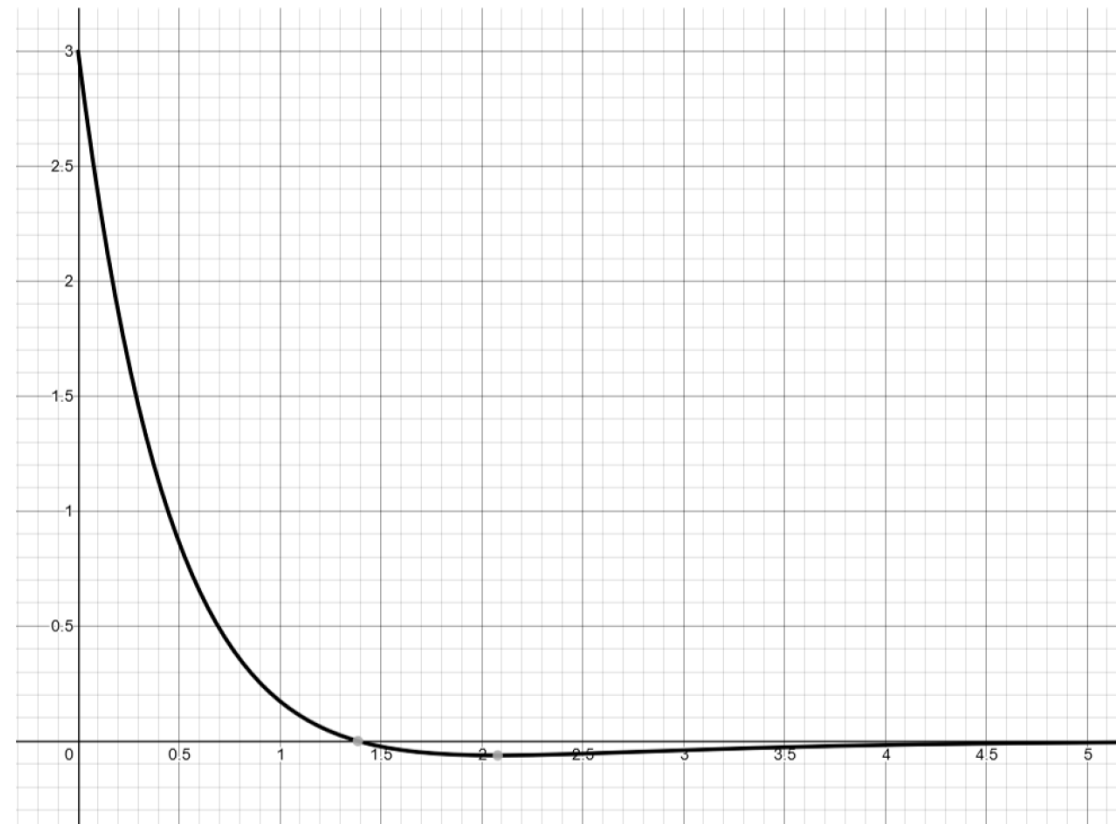
and the general solution is: $y = e^{-\alpha t}(A\cos(\omega t) + B\sin(\omega t))$. The sin and cos terms mean the solution will oscillate about the equilibrium, but the $e^{-\alpha t}$ term means it will tend to the equilibrium as time increases (remember $\alpha = c/2m$ is always positive)

To do Week 3:

- Watch ENGE702 lecture 6 pre-class recording on Canvas
- Watch ENGE702 lecture 7 pre-class recording on Canvas
- Watch ENGE702 lecture 8 pre-class recording on Canvas
- Complete Quiz 3
- Read Textbook Advanced Engineering Textbook 2.1, 2.2 & 2.4
- Review tutorial problems before this week's tutorial
- Matlab- 'MATLAB Onramp' 'Solving Ordinary Differential Equations With MATLAB'

Case 1: Overdamping continued

It is still possible that the mass passes equilibrium in this case, if the initial velocity is large enough.

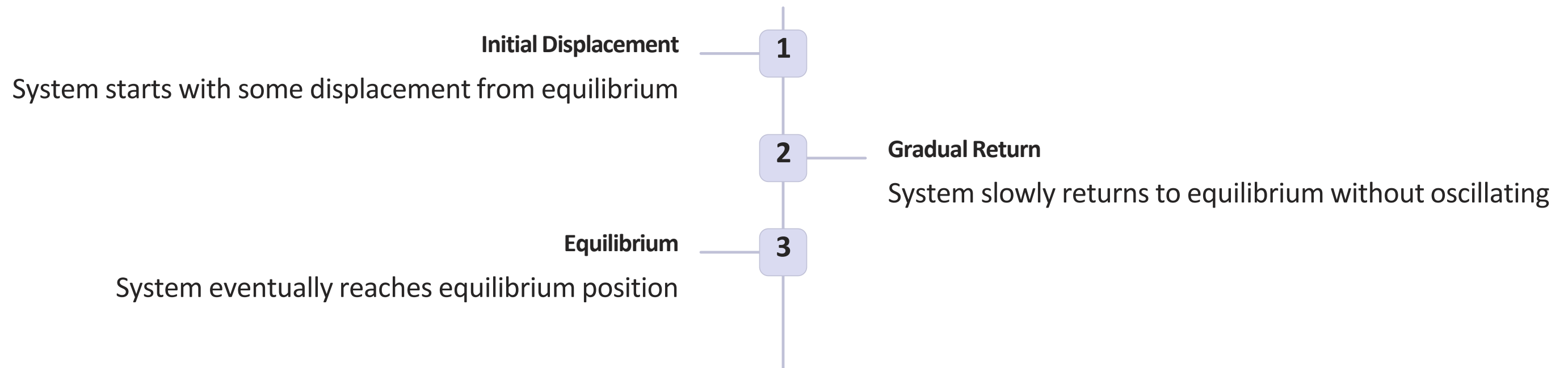
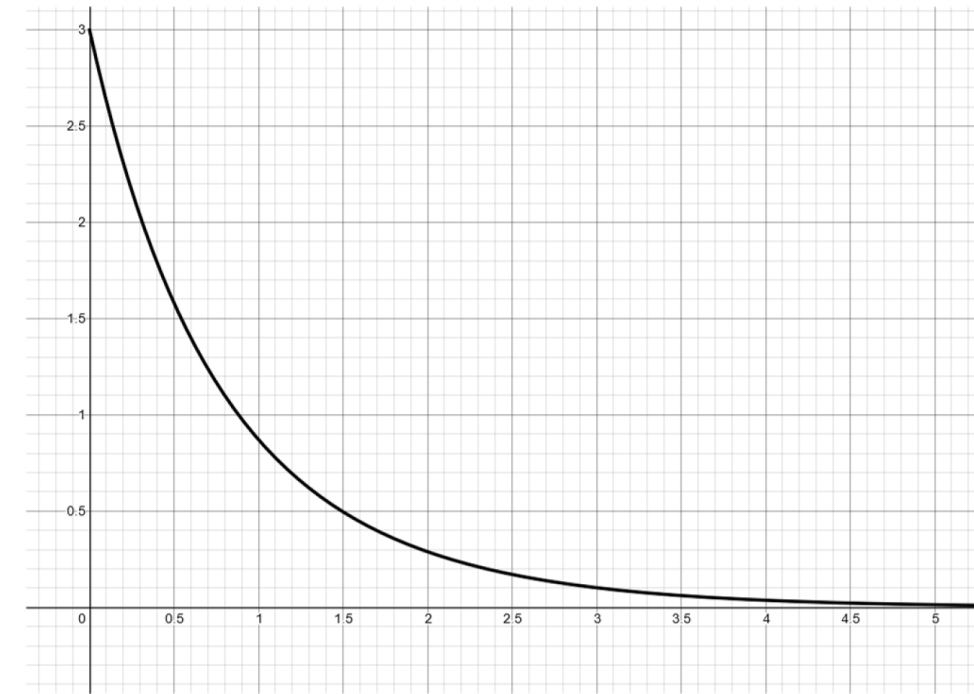


Case 1: Overdamping

Here the general solution is $y = c_1 e^{-(\alpha-\beta)t} + c_2 e^{-(\alpha+\beta)t}$

Note that both exponents are negative due to the nature of α and β , so the function will tend to 0 as time increases.

It is still possible that the mass passes equilibrium in this case, if the initial velocity is large enough.

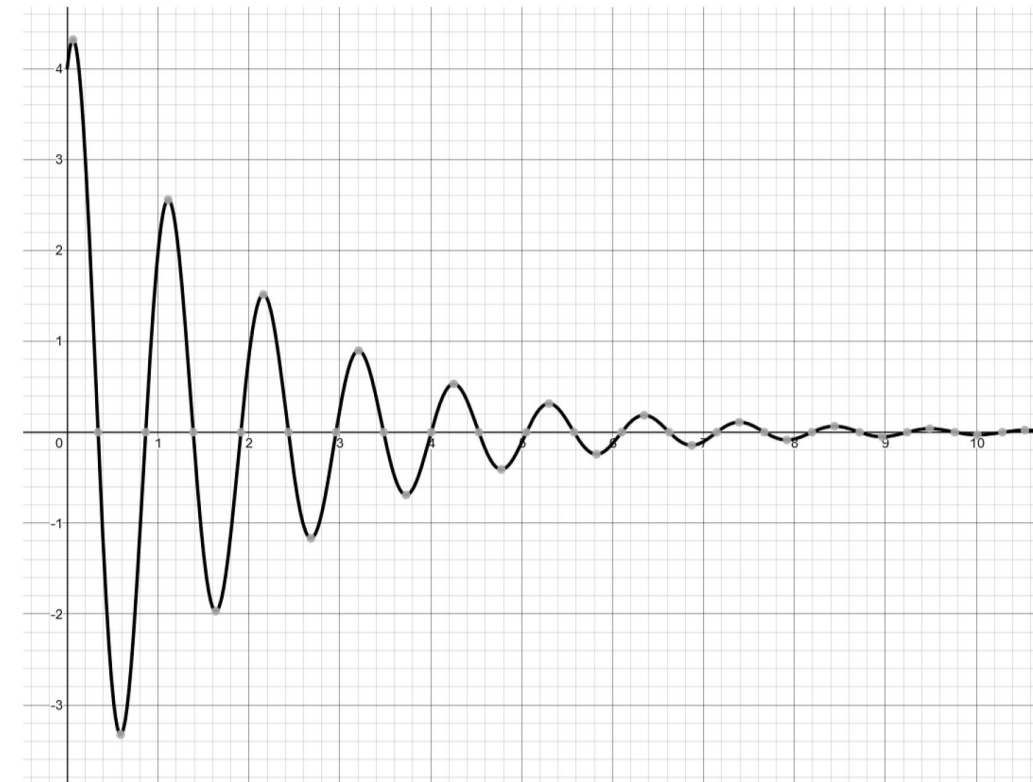


Case 3: Underdamping

This is the most complicated case. Here our solutions are $\lambda_1 = -\alpha + i\omega$ and $\lambda_2 = -\alpha - i\omega$, and the general solution is: $y = e^{(-\alpha t)}(A\cos(\omega t) + B\sin(\omega t))$.

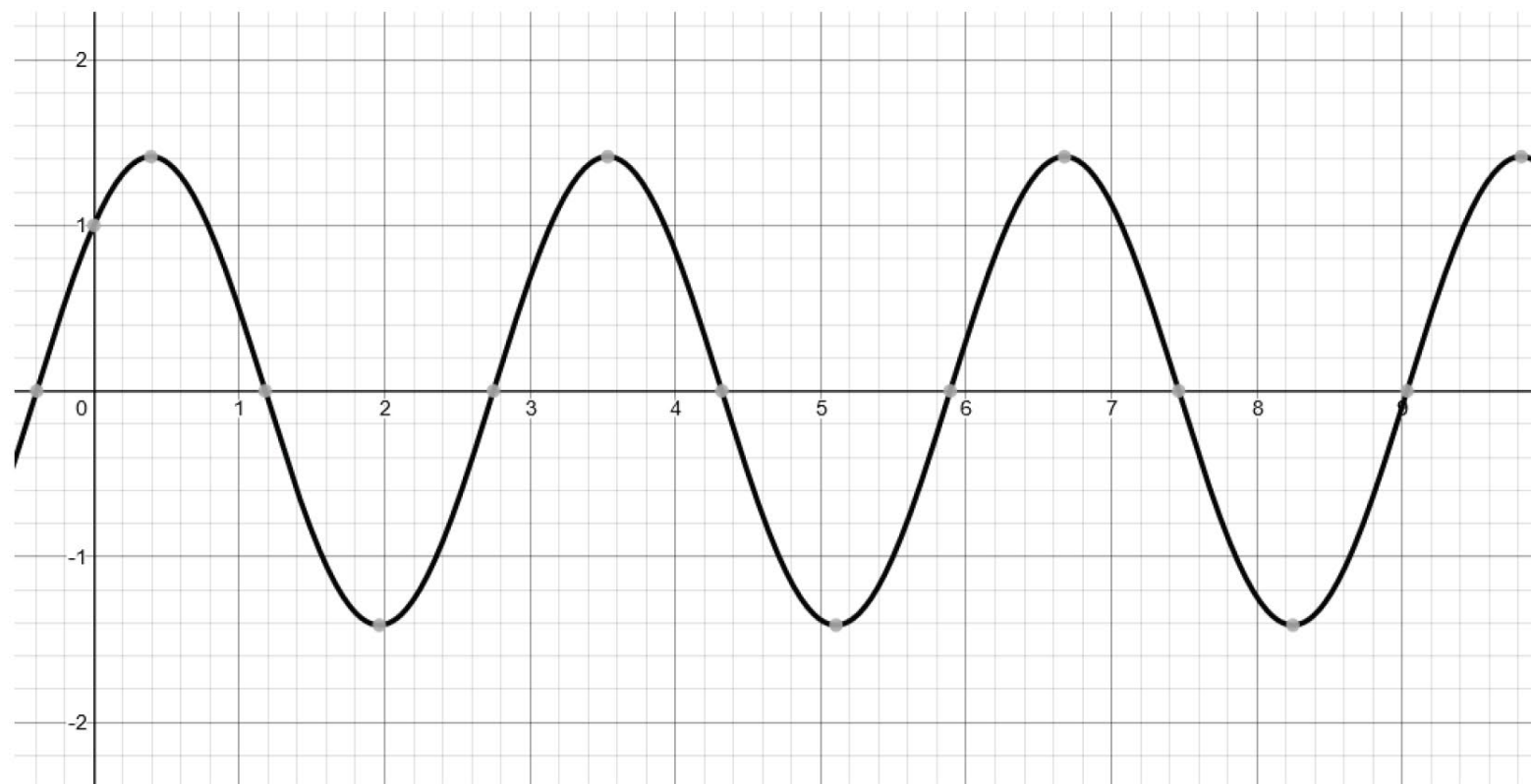
The sin and cos terms mean the solution will oscillate about about the equilibrium, but the $e^{(-\alpha t)}$ term means it will tend to the equilibrium as time increases (remember $\alpha = c/2m$ is always positive)

The solution will look something like this:



$$y = \cos(2t) + \sin(2t).$$

Plotting this gives:



Changing initial conditions will alter the amplitude and phase, but the frequency ω will always stay the same.



Exercises and References

Exercise

Suppose we have a mass - spring - damper system with a mass of $m = 1$, and a damping constant of $c = 4$. Find a value of the of the spring constant k that will give:

- underdamped motion
- critically damped motion
- overdamped motion

Text Book References

Chapter 2.4

Bryan (2022) 4.1 – 4.5



What did you get out of today?

- **Purpose of today's lecture?** What was the purpose of of today's lecture?
- **Key learning:** What is one key thing you learnt?
- **Questions:** Is there anything that is still puzzling you?

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